

Architecture Decision Record: ADR-004

Implementing AI-to-AI Communication over Chat

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Status: Accepted
Project: DVerse Communication Platform 2.0

1 Context

The DVerse platform includes a chat application interface through which users interact with the system. As AI agents become active participants in the platform, some agent-to-agent interactions should be visible in chat so that users can observe, inspect, and participate in multi-agent workflows.

This chat-based communication is distinct from Zenoh-based system messaging. Zenoh remains the dedicated asynchronous messaging layer, while the chat application provides a user-visible surface for selected agent interactions.

2 Decision

Sprint 4 will implement AI-agent-to-AI-agent communication through the chat application. Agents will be able to appear as identifiable participants in chat conversations, respond to each other, and preserve conversation context across multi-agent exchanges.

This implementation will include the following capabilities:

- Displaying agent messages in the chat application interface.
- Allowing agents to respond to one another inside chat conversations.
- Distinguishing agent-authored messages from user-authored messages.
- Preserving conversation context across multi-agent exchanges.
- Emitting OpenTelemetry traces and Pydantic Logfire logs for chat-mediated agent interactions.

Chat-visible events should reference the underlying Zenoh message identifiers or trace identifiers where appropriate. This allows engineers to move from a chat event to the corresponding distributed trace and structured logs.

3 Alternative Decisions

One alternative was to keep all agent-to-agent communication hidden from the chat interface. This was rejected because it would make multi-agent workflows less transparent to users.

Another alternative was to use chat as the primary system transport for agent-to-agent communication. This was rejected because chat is a user-facing interaction layer, while Zenoh is the dedicated messaging infrastructure for decoupled system communication.

4 Consequences

4.1 Positive

- Users can observe AI agents communicating with each other inside the chat interface.
- Agent-authored messages become distinguishable from user-authored messages.
- Chat interactions can be correlated with traces, logs, and underlying Zenoh events.
- Multi-agent workflows become easier to understand and inspect.

4.2 Negative

- Chat-mediated agent communication introduces additional requirements around message authorship and display clarity.
- Conversation context must be preserved carefully across multi-agent exchanges.
- The team must avoid treating chat as a replacement for the dedicated Zenoh messaging layer.

5 Scope

Deliverable	Description
Agent chat display	Show AI-agent messages in the chat application interface.
Agent replies	Allow AI agents to respond to each other inside chat conversations.
Message authorship	Distinguish agent-authored messages from user-authored messages.
Conversation context	Preserve context across multi-agent chat exchanges.
Observability links	Correlate chat events with traces, logs, and Zenoh message identifiers.

Statement on AI Use

Artificial intelligence was used to support the drafting of parts of this document. The generated text was subsequently reviewed, checked, and edited where necessary by me, the author, Abel-Raul Mazilu.